

Continually learning central pattern generator parameters for quadrupedal robot locomotion with active inference agents

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Abstract—Central pattern generators (CPGs) produce efficient rhythmic locomotion from a small set of parameters, but a set tuned for one terrain is no longer optimal once the terrain changes. We show that the CPG parameters yielding stable walking on flat ground and on a 10° incline are distinct, and that on a corridor of repeated transitions keeping the flat-optimal set more than doubles the fall rate relative to the terrain-matched optimum, so that resuming stable locomotion requires online adaptation. We propose an event-triggered scheme in which a prediction-error monitor detects each terrain change and re-tunes the CPG within the same walking bout, and we compare grid search, Bayesian optimization, and an autoregressive active inference agent that minimizes expected free energy. Grid search and Bayesian optimization destabilize the robot while probing candidate gaits, roughly doubling the fall rate, whereas the active inference agent selects parameters from a learned model without such trials and avoids this harm. Aligning its goal prior with the stability criterion, rather than with velocity tracking, makes it significantly outperform keeping the incumbent gait on a sustained transition, with the information-seeking term of the free energy essential to this gain; generalizing the improvement across repeated transitions remains open.

I. INTRODUCTION

Deploying legged machines in constantly changing environments demands an adaptability that engineers have long sought by replicating animal morphologies, such as snake-like robots for constrained spaces or quadruped platforms for irregular terrain [8], [4]. These platforms inherit the mechanical advantages of their biological counterparts, but their control systems often do not: the Boston Dynamics quadruped Spot, for instance, relies on centralized, sensor-intensive controllers that are effective yet complex and expensive, limiting wider deployment [4].

Central pattern generators (CPGs) offer a simpler alternative. In vertebrates, these spinal neural circuits generate the periodic, coordinated signals of movements such as walking, freeing higher brain centers to modulate speed and gait through simple inputs while sensory feedback keeps the rhythm coupled to the body and environment [5], [3]. This decentralized architecture reduces command latency and bandwidth and simplifies high-level control [6], making CPG-based controllers less energy intensive, simpler, and more affordable.

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A. Related Work

Among CPG models used in robotics, coupled nonlinear oscillators are biologically plausible and computationally efficient [2], [6]. Their mutual inhibition synchronizes the joints into coordinated gaits with smooth transitions between them [2], [11], [?], and their limit-cycle behavior yields self-sustaining oscillations that recover quickly from perturbations [11]. Governed by only a few parameters such as frequency and amplitude, they suit embedded systems with limited resources, yet most implementations still target high-end platforms, leaving their use on compact systems underexplored.

A complementary line of work tunes gait parameters as black-box optimization, from measured performance rather than a model of the dynamics. Bayesian optimization suits this setting, where each walking trial is expensive: it fits a Gaussian-process surrogate to the parameter-score pairs seen so far and proposes the next parameters by maximizing an acquisition function such as the upper confidence bound. Run *online*, it scores the parameters of the most recent window, augments the surrogate, and proposes the next window's parameters, refining the gait during walking; a trust region around the current best keeps proposals from jumping to destabilizing gaits [?]. We adopt this online Bayesian optimization as an adaptation baseline in Section IV.

In the following, we contribute

- the design of an intelligent agent that continuously updates parameters of a central pattern generator, and
- by benchmarking this agent against offline grid search and online Bayesian optimization.

II. PROBLEM STATEMENT

We first describe the base platform whose motion defines the locomotion objective, and then the controller driving it: a central pattern generator that produces the coordinated leg motion, augmented with an attitude virtual model controller that reactively corrects posture deviations. We then argue that, even with this reactive posture loop, the parameters yielding stable locomotion depend on the terrain, so that a parameter set tuned for one terrain degrades once the terrain changes. This motivates online adaptation of the parameters as a requirement for resuming stable locomotion.

A. Base platform and locomotion objective

The robot is a floating base actuated through its legs. We describe the trunk by its base state

$$\xi = (p, \Psi, v, \omega), \quad (1)$$

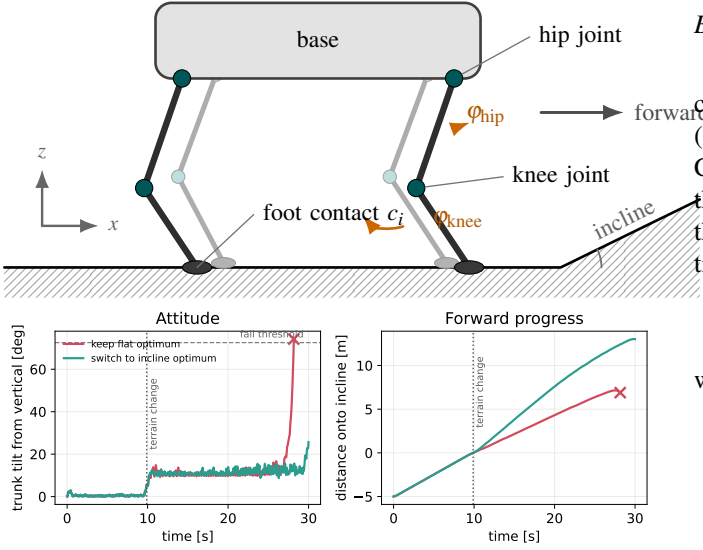


Fig. 1. Keeping the flat-optimal parameters after a terrain change destabilizes the robot. *Left*: the trunk tilt from vertical stays small on flat ground; after the transition (dotted line) both sets settle near the incline baseline, but keeping the flat optimum lets the tilt run away past the fall threshold (dashed line, $\times$), whereas switching to the incline optimum holds it near the baseline. *Right*: the robot climbs steadily after switching, whereas keeping the flat-optimal parameters ends in a fall and halted progress. Curves are one representative seed.

collecting the linear position $p \in \mathbb{R}^3$ and the angular position (orientation) $\Psi \in SO(3)$ of the trunk in a world frame, together with the linear velocity $v = \dot{p}$ and the angular velocity ω . The orientation Ψ is parameterized by the roll, pitch, and yaw (ψ_r, ψ_p, ψ_y) , of which the roll and pitch measure the deviation from an upright attitude.

The legs couple this base state to the joints. For a given motion of the base there is a corresponding configuration of the twelve joints, their rates, and the torques required to realize it,

$$(\varphi, \dot{\varphi}, \tau) = f(\xi), \quad (2)$$

where φ are the joint angles, $\dot{\varphi}$ their rates, and τ the joint torques. The map f is unknown: it depends on the full rigid-body dynamics, the terrain, and the switching contact between the feet and the ground, none of which are available in closed form.

Quadrupedal locomotion is then the problem of finding the joint torques τ such that the base state follows a desired behavior. Concretely, we require the trunk to maintain a target forward velocity v^* while holding an upright angular position, that is roll and pitch near zero,

$$\tau^* = \arg \min_{\tau} \|v - v^*\| + \|(\psi_r, \psi_p)\|. \quad (3)$$

Because f is unknown, these torques cannot be solved for from a model directly. Instead, a central pattern generator produces rhythmic joint-angle references, an attitude virtual model controller adjusts these references to counter posture deviations, and joint-level position controllers track the result through torques, shaping the base motion toward the objective. We introduce both components next.

B. Central Pattern Generator

The CPGs are modeled as a network of four mutually coupled Hopf oscillators, each driving one limb: left front (LF), right front (RF), left hind (LH), and right hind (RH). Coordination between limbs arises from inhibitory coupling through a gait-specific coupling matrix $K = [k_{ij}]$. Let ω_i be the balance between stance and swing frequencies, operationalized as

$$\omega_i = \omega_{\text{stance}} \zeta(-by_i) + \omega_{\text{swing}} \zeta(by_i), \quad (4)$$

with $\zeta(\cdot)$ a sigmoid function. Then, the oscillators evolve as

$$\dot{x}_i = \alpha(u - r_i^2)x_i - \omega_i y_i, \quad (5)$$

$$\dot{y}_i = \beta(u - r_i^2)y_i + \omega_i x_i + \gamma \sum_{j=1}^4 k_{ij} y_j + s_i, \quad (6)$$

where $r_i = \sqrt{x_i^2 + y_i^2}$ is the instantaneous amplitude, u defines the squared amplitude of the limit cycle, and $\alpha, \beta > 0$ are convergence rates (typically $\beta \gg \alpha$ to enforce stronger convergence along the y -axis). The term ω_i of Eq. 4 transitions between the stance and swing frequencies depending on the phase y_i : the sigmoid ζ smoothly interpolates between the two frequencies according to the sign and magnitude of y_i , with the parameter b setting the sharpness of this transition. The coupling gain γ strengthens the coordination between the limbs and reduces the dependence on initial conditions; the coupling matrix $K = [k_{ij}]$ is gait-specific.

The term s_i closes a feedback loop on foot-contact timing [?]. Leg i is classified as swinging when y_i exceeds a small positive threshold and as in stance below a small negative one, with a hysteresis band in between. Comparing this phase against the binary contact state of foot i yields

$$s_i = \begin{cases} K_{\text{stop}}(\omega_i x_i - \gamma \sum_j k_{ij} y_j) & \text{contact matches phase} \\ F_{\text{fast}} \text{sign}(y_i) & \text{otherwise} \end{cases} \quad (7)$$

and $s_i = 0$ inside the hysteresis band. The contact matches the phase when the foot is airborne during swing or loaded during stance; it contradicts the phase upon an early touchdown or an unexpected loss of ground contact. The gains K_{stop} and F_{fast} set how strongly the sensed footfalls reshape the nominal rhythm, synchronizing the oscillators to the actual contact events.

The oscillator states map to joint-angle references,

$$\varphi_i^{\text{hip}} = \varphi_0^{\text{hip}} + A_{\text{hip}} x_i, \quad \varphi_i^{\text{knee}} = \varphi_0^{\text{knee}} - A_{\text{knee}} \max(0, y_i) + \Delta_i, \quad (8)$$

where $\varphi_0^{\text{hip}}, \varphi_0^{\text{knee}}$ are the joint offsets of the standing pose and $A_{\text{hip}}, A_{\text{knee}}$ are the swing amplitudes. During swing ($y_i > 0$) the knee flexes to lift the foot while the hip advances with x_i ; during stance the knee holds its nominal extension. The term Δ_i is the posture correction supplied by the virtual model controller of Section II-C, and is zero for the open-loop CPG. Joint-level position controllers track these references, producing the joint torques τ of Eq. 2.

The behavior of the CPG is governed by a small set of parameters, collected in the vector

$$\theta = [\gamma, \omega_{\text{swing}}, \omega_{\text{stance}}, F_{\text{fast}}, K_{\text{stop}}, A_{\text{hip}}, A_{\text{knee}}, b], \quad (9)$$

comprising the coupling gain, the phase-dependent stance and swing frequencies, the fast-contact gain F_{fast} and stop gain K_{stop} of the contact feedback (Eq. 7), the hip and knee amplitudes of Eq. 8, and the sigmoid sharpness b of Eq. 4. These parameters set the gait shape, and are held fixed during locomotion.

C. Attitude Virtual Model Control

The contact feedback of Eq. 7 synchronizes the rhythm to the footfalls, but no part of the controller so far senses the trunk attitude. Yet roll and pitch deviations are precisely what drive the robot toward a fall (Eq. 3): on uneven or sloped ground the rhythmic pattern induces unwanted body rotations that it cannot itself correct. Virtual Model Control (VMC) addresses this by imagining virtual components, such as springs and dampers, attached to the robot and computing the joint actions that would arise if these components physically existed [9]. Ajallooeian et al. [1] augment a quadruped CPG with virtual springs spanning a trunk-fixed plane and a horizontal reference plane, so that the resulting virtual forces at the stance feet continually pull the trunk level.

We adopt this construction in a kinematic form suited to position-controlled joints. A virtual spring-damper on each attitude angle produces the correction commands

$$c_r = -(\kappa_p^r \psi_r + \kappa_d^r \dot{\psi}_r), \quad (10)$$

$$c_p = -(\kappa_p^p (\psi_p - \bar{\psi}_p) + \kappa_d^p \dot{\psi}_p), \quad (11)$$

where the κ 's are stiffness and damping gains. The roll is regulated toward zero, since straight walking involves no sustained bank and lateral tip-over is the dominant fall mode. The pitch is regulated toward a slowly adapting baseline $\bar{\psi}_p$, an exponential moving average of the recent pitch with a time constant on the order of one second, so that transient rocking is damped while the sustained pitch offset imposed by an incline is tolerated rather than fought. The commands are distributed over the legs as the knee corrections of Eq. 8,

$$\Delta_i = \text{clip}(\sigma_i^F c_p + \sigma_i^L c_r, \pm \delta_{\text{max}}), \quad (12)$$

where $\sigma_i^F = +1$ for the front legs and -1 for the hind legs, $\sigma_i^L = +1$ for the left legs and -1 for the right legs, and δ_{max} bounds the correction. Adjusting a knee angle changes the effective leg length, so Eq. 12 realizes the virtual springs of Ajallooeian et al. [1] through the position-controlled knees: the legs on a sinking side of the trunk extend to push that corner back up, while the legs on the rising side shorten.

The VMC gains are held at fixed, hand-tuned values and are not part of the parameter vector θ . The attitude loop thus acts at a fast timescale, correcting momentary posture deviations within a stride, but it leaves the rhythmic pattern itself untouched. Ajallooeian et al. [1] observe that such momentary corrections lose effectiveness as the terrain becomes more difficult, when the attitude corrections begin

to work against the forward progression planned by the CPG. Whether locomotion remains stable therefore still hinges on the gait parameters θ matching the terrain, which we examine next.

D. Terrain Dependence of the Optimal Parameters

For a fixed terrain, the parameter vector θ can be tuned to produce stable locomotion. Let e denote the terrain. Running the controller with parameters θ on terrain e , with the attitude feedback of Section II-C active, induces a distribution $p(\xi | \theta, e)$ over the base state, which we score by the locomotion objective of Eq. 3,

$$V(\theta, e) = -\mathbb{E}_{p(\xi | \theta, e)} \left[\frac{1}{v^*} \|v - v^*\| + \frac{1}{\psi_0} \|(\psi_r, \psi_p)\| \right], \quad (13)$$

setting $V = -c_f$ whenever the robot falls. The scale ψ_0 lets the attitude term dominate, encoding that remaining upright is the primary requirement. In practice the expectation is a time average over an evaluation window, computed on low-pass-filtered angles so that the gait's rhythmic rocking is not penalized, with the pitch measured relative to the terrain incline and progress beyond the target speed not rewarded. The best gait for terrain e is then the terrain-conditioned optimum

$$\theta^*(e) = \arg \max_{\theta} V(\theta, e). \quad (14)$$

The difficulty is that $\theta^*(e)$ is not constant across terrains. We consider two terrains, flat ground and a 10° incline, and maximize Eq. 13 on each with Bayesian optimization. The resulting optima differ markedly: relative to the flat optimum, the incline optimum pushes the coupling gain γ , both frequencies, the fast-contact gain F_{fast} , and the transition sharpness b to the top of their ranges while roughly halving the stop gain K_{stop} ; only the joint amplitudes are unchanged. The incline gait couples the limbs more strongly, swings the legs faster, and switches more sharply between stance and swing, and each parameter set is more stable on the terrain it was tuned for. The cost of the mismatch grows with exposure: on a single transition it is transient, roughly doubling the early fall rate before survivors settle, but on a corridor of repeated 10° transitions keeping the flat optimum leads to falls in 26% of runs against 11% with the incline optimum, a gap that holds over the whole traverse (Section IV).

Because θ is held fixed, a change in terrain leaves the controller at a now-suboptimal point $\theta^*(e_{\text{old}}) \neq \theta^*(e_{\text{new}})$, and the resulting attitude deviations compound: as the trunk tilts away from balance, recovery demands corrections beyond what the bounded attitude feedback of Eq. 12 can supply, and locomotion can fail before the new terrain is traversed. Fig. 1 shows a representative bout: after the transition, holding the flat-optimal parameters lets the tilt run away past the fall threshold and the robot topples, whereas switching to the incline-optimal parameters keeps the tilt near the incline baseline and the robot climbs steadily. Resuming stable locomotion after a terrain change therefore requires detecting that the terrain has changed and re-tuning θ toward $\theta^*(e_{\text{new}})$ online, quickly enough to preserve balance.

III. METHODS

A. Autoregressive active inference agent

The agent models the robot as a multivariate autoregressive process with exogenous inputs (MARX) [7]. The observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ is a linear-Gaussian function of a regressor $x_k = [u_k, \bar{u}_k, \bar{y}_k]^\top$ that stacks the current control $u_k = \theta$ with short memories $\bar{u}_k, \bar{y}_k$ of past inputs and outputs,

$$p(y_k | \Theta, x_k) = \mathcal{N}(y_k | A^\top x_k, W^{-1}), \quad (15)$$

with parameters $\Theta = (A, W)$. A matrix-normal-Wishart prior is conjugate to this likelihood [10, ID: P323], so the belief over Θ is updated in closed form by rank-one Bayesian filtering as each pair (x_k, y_k) arrives, and marginalizing it gives a Student-t posterior predictive whose mean and covariance,

$$\mu_k(u) = M_{k-1}^\top x_k, \quad \Sigma_k(u) = \frac{1 + x_k^\top \Lambda_{k-1}^{-1} x_k}{v_{k-1} - D_y - 1} \Omega_{k-1}, \quad (16)$$

are smooth functions of the control u through the regressor x_k . Here M, Λ, Ω, v are the filtered posterior parameters and $D_y = \dim y$; we take the recursions as standard for this conjugate model [7], [10] and keep the focus on the control objective.

The agent selects controls by minimizing an expected free energy. Two priors close the model: a control prior $p(u) = \mathcal{N}(u | \mu_0, \Upsilon^{-1})$ regularizes the parameters toward a reference gait μ_0 with precision Υ , and a goal prior $\bar{p}(y) = \mathcal{N}(y | m_*, S_*)$ encodes the target observation, the desired forward velocity v^* at an upright attitude, with S_* setting how tightly each channel is enforced. Planning over a horizon $\tau = \{k+1, \dots, k+T\}$ unrolls the model, substituting predictive means for the not-yet-observed outputs, and minimizes the cumulative divergence of the predicted observations from the goal prior,

$$G(u_\tau) = \sum_{t \in \tau} D_{\text{KL}}[p(y_t | u_t, \mathcal{D}_k) \| \bar{p}(y_t)]. \quad (17)$$

Expanding the divergence and adding the control prior gives the objective minimized at each planning step,

$$\begin{aligned} \hat{u}_\tau = \arg \min_{u_\tau} \sum_{t \in \tau} & \left[\underbrace{\frac{1}{2} \text{tr}(S_*^{-1} \Sigma_t(u_t)) - \frac{1}{2} \ln |\Sigma_t(u_t)|}_{\text{information gain}} \right. \\ & + \underbrace{\frac{1}{2} (\mu_t(u_t) - m_*)^\top S_*^{-1} (\mu_t(u_t) - m_*)}_{\text{goal seeking}} \\ & \left. + \underbrace{\frac{1}{2} (u_t - \mu_0)^\top \Upsilon (u_t - \mu_0)}_{\text{control cost}} \right]. \end{aligned} \quad (18)$$

The information-gain terms, through the predictive covariance Σ_t , drive the agent to resolve model uncertainty; the goal-seeking term, a Mahalanobis distance of the predicted mean μ_t from the target m_* , drives it toward stable walking; and the control cost keeps it near the reference gait.

B. Prediction-error triggering

Re-tuning θ at every step is wasteful and, as Section IV shows, actively destabilizing, so the agent adapts only when the incumbent gait is no longer matched to the terrain. We detect that event by testing, online, whether the incumbent controls still minimize the control objective of Eq. 18.

Consider the one-step objective $G(u)$ under two simplifications that hold in the monitoring regime, once the gait is steady and the belief identified. The control cost is a fixed price rather than a signal, and the information-gain term is flat in u (its only dependence is through $\ln(1 + x^\top \Lambda^{-1} x)$, negligible for large Λ). What remains is the goal cross-entropy

$$G(u) = \frac{1}{2} \text{tr}(S_*^{-1} \Sigma) + \frac{1}{2} (\mu(u) - m_*)^\top S_*^{-1} (\mu(u) - m_*), \quad (19)$$

with $\mu(u) = M^\top x$ affine in u through the learned control-to-output gain $B := \partial \mu / \partial u$, a block of $M^\top$. The incumbent θ remains terrain-matched precisely when it is a stationary point of G . Because μ is affine in u and G is quadratic in μ , G is exactly convex-quadratic in u , with gradient and Hessian

$$g = \nabla_u G(\theta) = B^\top S_*^{-1} (\mu - m_*), \quad H = \nabla_u^2 G = B^\top S_*^{-1} B. \quad (20)$$

The value $G(\theta)$ alone does not test optimality: the goal error $\mu - m_*$ can grow along directions the controls cannot affect, terrain error that B annihilates, leaving θ stationary despite a high cross-entropy. The gradient separates the fixable error from the rest. Its affine-invariant magnitude is the Newton decrement

$$\lambda^2 = g^\top H^{-1} g = \|\Pi_{\mathcal{R}(B)}(\mu - m_*)\|_{S_*^{-1}}^2, \quad (21)$$

the squared goal error projected onto the controllable subspace $\mathcal{R}(B)$. Since G is exactly quadratic, $\frac{1}{2} \lambda^2 = G(\theta) - \min_u G(u)$ is not a local approximation but the exact sub-optimality gap, the cross-entropy a full re-tuning of the gait would recover. Triggering on λ^2 therefore fires on *fixable* mismatch rather than on raw prediction error, and correctly stays silent when the error lies in the uncontrollable subspace, where no gait switch would help.

The threshold follows from the same statistic. Under the null hypothesis that the incumbent is still optimal, the goal error is predictive noise, $\mu - m_* \sim \mathcal{N}(0, \Sigma)$, so λ^2 is a quadratic form of a Gaussian and hence (generalized) χ^2 with $r = \text{rank}(B) \leq \min(D_u, D_y)$ degrees of freedom. A target false-alarm rate α then fixes

$$K = \chi_r^2(1 - \alpha), \quad (22)$$

a threshold with units and a calibrated meaning in place of a hand-tuned level. Equivalently, restoring the control cost as the price of adapting gives a decision-theoretic rule: trigger only when the recoverable gain outweighs the cost of the corrective step $\Delta u = -H^{-1} g$, that is $\frac{1}{2} \lambda^2 > \frac{1}{2} \Delta u^\top \Upsilon \Delta u$, so that adaptation fires exactly when the full expected free energy would decrease.

In practice, calibrating the null scale of λ^2 on a short flat-walking prefix reduces the test to a dimensionless ratio

near unity during nominal locomotion, the form used in Section IV. The attitude channels of S_* are tightened relative to the velocity channels so the statistic responds to a loss of balance rather than to the slower climb up the incline, and it is smoothed by an exponential moving average to reject stride-frequency ripple. An event fires when the smoothed statistic exceeds K for P consecutive steps; the monitor then disarms and re-arms only once it falls back well below K , so a single terrain change produces one event while a corridor of repeated changes re-triggers once per transition. Symmetrically, adaptation is paused whenever the statistic returns below K , so re-tuning runs only while the incumbent is provably suboptimal.

IV. EXPERIMENTS

We evaluate three parameter-selection strategies for the Hopf-oscillator CPG controller on the Laikago quadruped in PyBullet: (i) an offline *grid search* over a small set of CPG parameters; (ii) online *Bayesian optimization* (BO) with a Gaussian-process surrogate and a UCB acquisition function; and (iii) the proposed *MARX-EFE* active inference agent, which continually updates CPG parameters by minimizing expected free energy. Grid search serves as a fixed, non-adaptive reference point, BO serves as a state-of-the-art black-box optimizer, and MARX-EFE is the proposed continual learning agent.

All three methods optimize in the same 8-dimensional parameter space with identical bounds:

$$\theta = [\gamma, \omega_{\text{swing}}, \omega_{\text{stance}}, F_{\text{fast}}, K_{\text{stop}}, A_{\text{hip}}, A_{\text{knee}}, b],$$

corresponding to the coupling gain, the two phase-dependent frequencies, the Righetti-style fast-contact gain, the stop-gain, the hip and knee amplitudes, and the sigmoid sharpness b from Eq. 4. Each trial is a $T = 4.5$ s episode of simulated walking on flat ground with $\Delta t = 0.01$ s, a 1.5 s settling window excluded from the metrics, and a target forward velocity of 1.0 m/s. Parameter updates occur *online*: every trial carries the CPG state forward and smoothly interpolates between the previous and new parameters over the first 150 steps.

a) Performance metrics: We score each method on four quantities computed over the 3.0 s steady-state window: the scalar objective

$$J = w_1 R_v - w_2 \text{CoT} - w_3 S, \quad (23)$$

with velocity reward $R_v = \frac{1}{T} \sum_k \exp(-\|v_k - v^*\|^2 / \sigma_v^2)$, cost of transport $\text{CoT} = \int P_{\text{mech}} dt / (mgd)$, and combined RMS stability $S = \sqrt{\text{RMS}(\gamma^r)^2 + \text{RMS}(\gamma^p)^2}$; the forward distance d_y ; the lateral drift $|d_x|$; and the fall rate. To compare sample efficiency, we report the number of trials N_{walk} required to reach $J \geq J_{\text{th}}$ with cumulative fall rate below r_{th} , and the cumulative forward distance D_{cum} accumulated over the optimization horizon.

A. Experiment 1: Comparison of optimization strategies

In this experiment we compare grid search, BO, and MARX-EFE on flat terrain with a fixed target velocity of 1.0 m/s. Grid search is run offline on a restricted (α, β, b) sub-grid with the remaining parameters fixed, producing a ranking of static configurations. BO and MARX-EFE are run for 100 online trials, each repeated over 5 random seeds. For BO we use UCB with a decaying schedule $\beta_t = \max(\beta_{\min}, \beta_0 \gamma^{-t_0})$ and standardized outputs; for MARX-EFE we use a matrix-normal–Wishart prior over the 8D parameter regression coefficients and infer a new parameter vector every 50 simulation steps by minimizing the EFE of Eq. 18 with an L^2 effort regularizer.

Adaptation is triggered by the optimality monitor of Section III-B. A MARX belief (Eq. 15) is filtered online from the observation $y_k = (v_x, v_y, \psi_p, \psi_r)$ with forgetting factor 0.995, under a stability-oriented goal prior whose attitude channels are tightened, a 20° standard deviation on pitch to tolerate the steady incline offset and 6° on roll, so the statistic responds to a loss of balance rather than to the slower climb. At each step the monitor evaluates the Newton decrement λ^2 (Eq. 21), the goal cross-entropy that re-tuning the gait could recover, from the model’s steady-state control gain B . Because B is the multi-step feedthrough of the identified model, λ^2 reflects the sustained mismatch the incline creates, whereas a one-step innovation at 100 Hz is dominated by persistence and barely registers the change. An event fires when λ^2 exceeds the decision-theoretic budget of Section III-B, the control-prior price of a reference gait move of 2% of each parameter’s range, which for the monitor’s control-prior width evaluates to $\tau \approx 2.3$; unlike a normalized error ratio, this threshold carries units and needs no per-seed flat-walking baseline. On flat ground the incumbent is optimal and λ^2 stays below τ ; when the trunk crosses onto the incline the controllable part of the goal error grows and λ^2 crosses τ shortly after, firing once. For the repeated-transition corridor of Section ?? the monitor re-arms once λ^2 falls back below $\frac{1}{2}\tau$, so it fires once per ramp rather than latching.

B. Adaptation protocol

When the trigger fires, the selected method begins proposing CPG parameters, one candidate per 1.5 s window, ramped in over 0.3 s, and each window is scored by Eq. 13. The optimizers search a trust region around the incumbent and share a common safety layer: the best-scoring parameters seen so far are restored for one window whenever a candidate scores well below them. Without the safeguard, every method, MARX-EFE included, destabilizes the robot within seconds, so it is applied uniformly across methods. The trust-region radius, however, interacts with θ^* : at ± 0.25 of the parameter range the incline optimum is not reachable, because $\theta^*(10^\circ)$ raises the coupling gain γ across its full range and drops the sharpness b , so only 30% of the flat-to-incline shift lies inside the box. We therefore report both the tight ± 0.25 region, which equalizes caution across the black-box optimizers, and a full-range region in which $\theta^*(10^\circ)$ is

TABLE I

EVENT-TRIGGERED ADAPTATION ON THE FLAT-TO- 10° TRANSITION ($N = 100$ SEEDS). LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$. BEST ONLINE METHOD IN BOLD.

Metric	reference		online method		
	no adapt	oracle	grid	BO	MARX-EFE
Falls, 20 s [%]	40	47	83	90	39
Falls, 10 s [%]	19	8	37	37	10
Up time [s]	16.7	17.3	12.3	12.2	17.8
Mean $\bar{V}$	-0.36	-0.30	-0.54	-0.54	-0.33

TABLE II

ALIGNING THE GOAL PRIOR WITH STABILITY ON THE FLAT-TO- 10° TRANSITION ($N = 100$ SEEDS, TRUST REGION WIDENED SO $\theta^*(10^\circ)$ IS REACHABLE). MARX-EFE USES THE VELOCITY GOAL PRIOR; MARX-EFE-S TIGHTENS THE ATTITUDE TERM TO THE STABILITY SCALE; THE LAST COLUMN REMOVES THE EPISTEMIC TERM FROM MARX-EFE-S. LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$.

Metric	reference		MARX-EFE goal prior	
	no adapt	oracle	velocity	stability
Falls, 20 s [%]	40	47	44	27
Falls, 10 s [%]	19	8	17	7
Up time [s]	16.7	17.3	16.8	18.4
Mean $\bar{V}$	-0.36	-0.30	-0.37	-0.29

reachable, which is required to ask whether a method can adapt toward it. MARX-EFE uses a forgetting factor of 0.995 so that its model tracks the change, and its control prior is centered on the current parameters. Its goal prior encodes the target observation; we compare the original prior, which rewards the target forward velocity with a loose attitude term, against a *stability-aligned* prior whose attitude term is tightened to the $\psi_0 = 10^\circ$ scale of the criterion, so that the expected-free-energy objective targets the attitude that θ^* optimizes.

All arms share an identical pre-trigger trajectory per seed so that the event fires at the same instant in every arm. Two references bracket the online methods: *no adaptation* keeps $\theta^*(\text{flat})$, a lower reference, and the *oracle* switches directly to $\theta^*(10^\circ)$, an upper reference that presumes the new optimum is known. The online methods are *grid*, *BO*, and the proposed *MARX-EFE*, the last with either the velocity or the stability-aligned goal prior; we write *MARX-EFE-S* for the stability-aligned variant. We report the fall rate over the 20 s after the trigger, the fall rate over the first 10 s (the transition phase, where the parameter mismatch itself is the hazard), the mean time the robot stays upright, and the mean windowed criterion $\bar{V}$.

C. Results

Table I and Fig. ?? summarize the single-transition comparison at the ± 0.25 trust region. The shift in the optimum matters at the transition: within the first 10 s the oracle falls in 8% of seeds against 19% for no adaptation (McNemar $p =$

0.03), so switching to terrain-matched parameters roughly halves the early falls.

Trial-based search is destructive when run online. Grid and BO must spend whole windows walking on speculative candidates, and on an incline a poor candidate does not merely score badly, it tips the robot over. Even confined to the trust region and protected by the safeguard, this roughly doubles the fall rate relative to not adapting (83% and 90% against 40%; McNemar $p \leq 4 \times 10^{-10}$, upright-time Wilcoxon $p \approx 10^{-12}$). MARX-EFE avoids these physical trials by selecting from its model, and stays far below the black-box optimizers (early 10% against 37%, $p < 10^{-5}$). Over the full horizon, however, it matches no adaptation (39% against 40%, not significant); with the velocity goal prior it avoids harm but does not improve on holding the incumbent.

The parameter trajectories explain why (Fig. ??). Under the velocity goal prior the selected $\theta(t)$ stays at $\theta^*(\text{flat})$: MARX-EFE does not move toward the incline optimum. Two factors are responsible. The tight trust region does not contain $\theta^*(10^\circ)$, so the incline optimum is unreachable; but widening it does not help either, and the deeper cause is the goal prior, which rewarded forward velocity with only a loose (45°) attitude term and so did not target the attitude stability that θ^* optimizes. Ablations confirm that the machinery was idle under this goal: removing the epistemic term (a greedy predictive selector), removing the forgetting factor, or removing the safeguard each leaves the outcome statistically unchanged, and re-centering the control prior away from the incumbent only destabilizes the robot. The safeguard is thus load-bearing for grid and BO, which wander, but not for MARX-EFE, which holds the incumbent regardless.

Aligning the goal prior with the stability criterion changes this. Table II reports the effect of tightening the attitude term to the $\psi_0 = 10^\circ$ scale (MARX-EFE-S), with the trust region widened so that $\theta^*(10^\circ)$ is reachable. MARX-EFE-S falls in 27% of runs over 20 s against 40% for no adaptation (McNemar $p = 0.04$) and in 7% against 19% over the first 10 s ($p = 0.008$), the first arm to improve significantly on holding the incumbent, and below the oracle as well ($p = 0.006$). It achieves this not by reaching $\theta^*(10^\circ)$, which stays out of range of its small adjustments, but by making minor attitude-directed corrections around the incumbent: a nearby, model-guided change suffices to keep the trunk upright. The epistemic term is now load-bearing: with it removed the same goal prior yields 42%, no better than no adaptation. The active-inference machinery therefore contributes, but only once its goal prior encodes the quantity being optimized.

A single incline is a mild test of adaptation, because a fixed gait that survives the transition can coast through the rest of the bout. We therefore repeat the experiment on a corridor of two consecutive 10° ramps separated by flat landings, so that the robot meets the terrain change more than once and a mismatched gait is punished repeatedly. Here the value of terrain-matched parameters is both larger and durable: the oracle falls in 11% of runs against 26% for no adaptation, and the gap holds over the whole traverse rather than washing

placeholder: cumulative fall rate and D_{cum}

Fig. 2. Safety and efficiency. (Left) cumulative fall rate vs. trial index; (right) cumulative forward distance D_{cum} walked across all trials, a measure of sample-efficient exploration.

TABLE III

REPEATED 10° TRANSITIONS (TWO RAMPS, $N = 100$ SEEDS, FULL-RANGE TRUST REGION). MARX-EFE-S IS THE STABILITY-ALIGNED AGENT. LOWER IS BETTER EXCEPT UPRIGHT TIME AND $\bar{V}$.

Metric	reference		online method		
	no adapt	oracle	grid	BO	MARX-EFE
Falls, 20 s [%]	26	11	87	92	23
Falls, 10 s [%]	9	10	65	73	11
Upright time [s]	17.4	18.7	9.6	9.0	17.7
Mean $\bar{V}$	0.22	0.44	-0.46	-0.53	0.27

out as it did on the single incline. The optimality monitor re-arms once λ^2 falls back below the budget after each ramp, detecting every transition without firing on the flat landings between them. Online black-box search remains destructive: grid and BO fall in 87% and 92% of runs at the full-range trust region (38% and 41% even when confined to the tight region), far above both references.

The model-based agent does not capture the oracle’s advantage here, however (Table III). With either goal prior it falls in 23–24% of runs, statistically indistinguishable from no adaptation (26%, McNemar $p = 0.83$) and well above the oracle (11%, $p = 0.007$). The stability-aligned prior that improved on the single sustained incline does not replicate that gain when the transitions recur, and the epistemic term confers no advantage in this setting either. A gait corrected for one ramp is not necessarily right for the next after an intervening flat stretch, and the small corrections that sufficed on a single incline do not reach the alternating optimum. Capturing the oracle’s durable benefit across repeated transitions, rather than only avoiding the destabilization that black-box search causes, therefore remains open.

D. Natural terrain

The transitions so far are between two known surfaces with pre-computed optima, which lets an oracle mark the achievable ceiling. A natural landscape offers no such reference: the ground changes in both friction and geometry many times along the path, and the parameter optimum drifts continuously, so there is no single terrain-matched gait to switch to. We therefore replace the two-surface corridor with a natural transect of forward bands of grass, gravel, rock,

TABLE IV

ONLINE ADAPTATION ON A NATURAL TRANSECT OF FRICTION AND GEOMETRY BANDS ($N = 10$ SAMPLED TRANSECTS, DECISION-THEORETIC TRIGGER). THERE IS NO ORACLE, BECAUSE THE OPTIMUM DRIFTS ALONG THE PATH. SURVIVAL DISTANCE AND BANDS CROSSED ARE HIGHER-IS-BETTER; FALL RATE AND TRUNK TILT ARE LOWER-IS-BETTER; $\bar{V}$ IS HIGHER-IS-BETTER. BEST ONLINE METHOD IN BOLD.

Metric	reference	online method		
	no adapt	grid	BO	MARX-EFE
Distance [m]	34.2	32.9	33.2	34.3
Falls [%]	50	60	60	40
Bands crossed	4.7	4.5	4.5	4.6
Trunk tilt [deg]	2.2	2.3	2.3	2.2
Mean $\bar{V}$	-0.27	-0.50	-0.43	-0.31

and shallow water in random order and length, each band carrying its own friction and its own micro-geometry: gentle undulation on grass, fine roughness on gravel, scattered low rocks to step over, and a slippery depression at the water. Bands are 6 to 10 m long, so each surface persists for 12 to 20 s of walking, and the robot walks a single continuous bout of up to 80 s rather than resetting after one transition. It starts on grass with θ^* (flat), whose friction $\mu \approx 0.7$ makes the flat optimum the natural incumbent, and adapts online whenever the optimality monitor fires. The monitor re-arms after each band as on the repeated-ramp corridor of Section IV-C, so a single run triggers once per transition it deems significant, on average 1.5 times over the bout. Because there is no oracle, and almost every gait eventually falls on a long enough transect, the discriminating metric is the *survival distance*, the distance walked before a fall or the end of the terrain. We report it alongside the fall rate, the number of bands crossed, the mean trunk tilt $\sqrt{\psi_r^2 + \psi_p^2}$ after the baseline window, and the mean windowed criterion $\bar{V}$, over $N = 10$ sampled transects.

The natural transect echoes the two-surface findings (Table IV). Survival distances overlap across all arms, from 33 to 34 m with a per-seed spread of about 11 m that reflects where the first fall happens, so distance alone does not separate the methods. The fall rate and the windowed criterion do, and they order the arms consistently. Online black-box search is again the most fragile: grid and BO fall in 60% of runs against 50% for no adaptation, and their mean criterion is the lowest (-0.50 and -0.43 against -0.27), because a speculative candidate walked on a low-friction or uneven band tips the robot just as it did on the incline. MARX-EFE is the only arm to improve on holding the incumbent, falling in 40% of runs while matching no adaptation on trunk tilt and criterion. This mirrors its behavior on the single and repeated inclines: it selects small model-guided corrections rather than committing whole windows to physical trials, which avoids the destabilization that black-box search causes. The small sample leaves these orderings short of statistical significance, but their direction matches the two-surface experiments and

holds once the bands are long enough to expose a mismatched gait for a sustained stretch.

E. Experiment 2: Sensitivity to exploration schedule

A known failure mode of BO is that its behaviour depends strongly on the acquisition-function exploration schedule. We sweep five β_t schedules (conservative, moderate, balanced, late_decay, exploratory) while holding all other BO settings fixed, and compare them against a single MARX-EFE run. The purpose is to show (a) that BO requires schedule tuning to achieve competitive sample efficiency, and (b) that MARX-EFE does not require such a schedule because the exploration–exploitation trade-off is implicit in the epistemic term of the expected free energy.

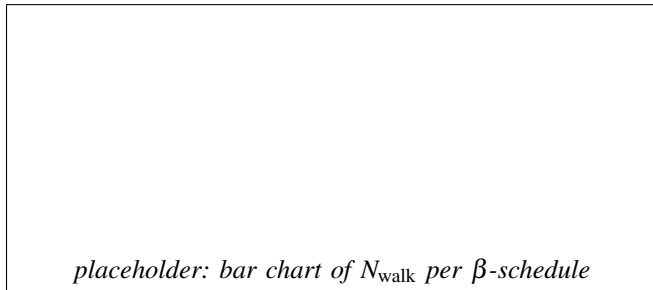


Fig. 3. BO sensitivity to the UCB β_t schedule. N_{walk} (lower is better) for each schedule, versus the MARX-EFE reference.

F. Experiment 3: Online adaptation to target-velocity changes

A key claim of the continually-learning agent is that it can adapt online to non-stationary tasks without re-tuning. We change the target velocity v^* mid-episode from 0.5 m/s to 1.0 m/s to 0.8 m/s, and measure how quickly each method tracks the new set-point. Only methods that update their parameters online (BO and MARX-EFE) are evaluated here.

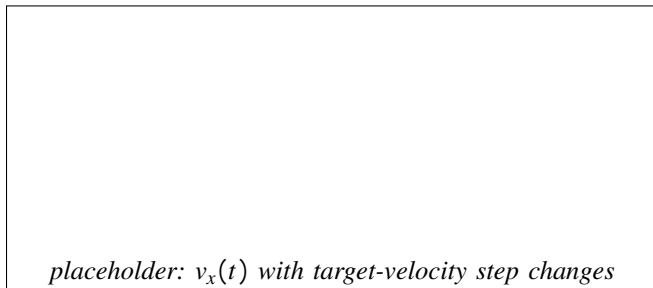


Fig. 4. Online target-velocity tracking. Realised forward velocity $v_x(t)$ under a step change in v^* ; BO and MARX-EFE shown separately.

V. DISCUSSION

Our results separate two aspects of online adaptation that are easily conflated: deciding *when* to change the gait and deciding *what* to change it to. The optimality trigger resolves the first cheaply and reliably, firing on the controllable part of the prediction error, once per terrain change and never on flat ground, and it re-arms to detect a sequence of transitions.

The second is where the methods diverge. Grid search and Bayesian optimization can only judge a candidate gait by walking on it, and on an incline a poor candidate does not merely score badly, it tips the robot over; even confined to a trust region and protected by a revert-to-best safeguard, this exploration roughly doubles the fall rate relative to not adapting at all. The active inference agent instead selects parameters from a model that it updates every control step, so it avoids these destabilizing trials. What the agent selects, however, depends critically on its goal prior. With a goal that tracks forward velocity, the agent simply holds the incumbent gait: it avoids harm but does not improve on not adapting, and its information-seeking term, forgetting factor, and revert-to-best safeguard are all idle. Once the goal prior encodes attitude stability, matching the criterion that defines the terrain optimum, the agent instead makes small model-guided corrections that significantly reduce falls on a sustained transition, and the information-seeking term becomes essential to that gain. The advantage is therefore not faster search, but the ability to improve the gait without risking it, and it appears only when the agent’s objective encodes the quantity being optimized.

Several limitations qualify these conclusions. The improvement does not yet generalize: on a corridor of repeated transitions, where the terrain-matched optimum confers a large and durable advantage, neither the stability-aligned agent nor any goal-prior setting we swept captures it, and the selected parameters never travel to the terrain optimum. The barrier appears structural rather than a matter of tuning: the linear autoregressive model supports small local corrections but cannot predict which distant parameter settings are stable, and loosening the control prior to permit larger moves only reintroduces destabilizing exploration. Robust adaptation across recurring terrain changes, for instance by re-selecting per transition or by a model expressive enough to navigate to the terrain optimum, is the central open problem. The study is also in simulation with reference optima computed offline, and the open-loop CPG has a narrow terrain envelope: beyond roughly 10° of incline, and on declines, no fixed parameter set keeps it upright, so controller fragility rather than the adaptation method limits generalization to steeper terrain. Coupling the agent to closed-loop attitude feedback and validating on hardware are natural next steps.

VI. CONCLUSIONS

APPENDIX

ACKNOWLEDGMENT

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